

I. SVD Fundamentals (1–10)

1. What is the main limitation of eigenvalue decomposition that SVD overcomes?

Answer: Eigenvalue decomposition requires a square matrix, while SVD can decompose **any rectangular matrix**.

2. What is the Singular Value Decomposition (SVD)?

Answer: SVD is a factorization of a matrix ($X_{n \times p}$) into three matrices:

$$X = U \Sigma V^H$$

where (U) and (V) are unitary, and (Σ) is diagonal with non-negative singular values.

3. Any rectangular matrix ($X_{n \times p}$) can be written as the product of which three matrices in the SVD, and what are their respective sizes?

Answer:

$$X = U_{n \times n} \Sigma_{n \times p} V_{p \times p}^H$$

4. What mathematical property must the unitary matrices (U) and (V) in SVD possess regarding their inverse and Hermitian transpose?

Answer: ($U^{-1} = U^H$) and ($V^{-1} = V^H$), meaning they are **unitary (or orthogonal if real)**.

5. What type of matrix is ($\Sigma_{n \times p}$) in the SVD?

Answer: A **rectangular diagonal matrix** with non-negative singular values on the diagonal.

6. The SVD formula ($X = U \Sigma V^H$) physically decomposes the linear operator (X) into what three types of operations?

Answer:

1. Rotation/reflection by (V^H),
2. Scaling by (Σ),
3. Rotation/reflection by (U).

7. In the context of SVD, what are the columns of the matrix (V) called?

Answer: Right singular vectors.

8. In the context of SVD, what are the columns of the matrix (U) called?

Answer: Left singular vectors.

9. Write the equation that defines a singular value (σ) in terms of the matrix (X) and its singular vectors.

Answer:

$$\begin{bmatrix} X v_i = \sigma_i u_i \quad \text{and} \quad X^H u_i = \sigma_i v_i \end{bmatrix}$$

10. What does the SVD relation ($X = U \Sigma V^H$) represent in terms of mapping a feature vector?

Answer: It maps an input vector (v_i) into a scaled output (u_i) along the principal directions defined by (X).

II. SVD and Eigen Decomposition Relationship (11–19)

11. What is the symbolic product of matrices that results in the eigenvector matrix for the square matrix ($X^H X$)?

Answer: ($X^H X = V \Sigma^H \Sigma V^H$), so columns of (V) are eigenvectors of ($X^H X$).

12. If (X) is a real matrix, what is the relationship between the singular values (σ) of (X) and the eigenvalues (λ) of ($X^T X$)?

Answer: ($\lambda_i = \sigma_i^2$)

13. The right singular vectors (V) of (X) are the eigenvectors of which square matrix?

Answer: ($X^H X$) (or ($X^T X$) if real).

14. The left singular vectors (U) of (X) are the eigenvectors of which square matrix?

Answer: ($X X^H$)

15. What is the symbolic equation that shows the eigenvectors and eigenvalues of ($X^H X$) in terms of the singular values and singular vectors of (X)?

Answer:

$$\begin{bmatrix} X^H X v_i = \sigma_i^2 v_i \end{bmatrix}$$

16. If (X) is a real symmetric square matrix, how does SVD relate to its standard eigendecomposition?

Answer: The SVD and eigendecomposition coincide, and ($U = V$), (Σ) contains absolute eigenvalues.

17. What is the fundamental difference in the mathematical process between SVD and PCA?

Answer: PCA is based on the eigendecomposition of the covariance matrix, whereas SVD decomposes the data matrix directly.

18. PCA relies on the eigendecomposition of the covariance matrix. If the data is normalized, what is the relationship between the covariance matrix and the matrix ($X^T X$)?

Answer: The covariance matrix is proportional to ($X^T X$):

$$\begin{aligned} &[\\ \text{Cov}(X) &= \frac{1}{n-1} X^T X \\ &] \end{aligned}$$

19. What is the condition for an eigendecomposition to be possible in terms of the eigenvectors of the matrix (A)?

Answer: (A) must be diagonalizable, i.e., it has a complete set of linearly independent eigenvectors.

III. SVD and PCA (20–25)

20. What do the right singular vectors (V) of the data matrix (X) directly correspond to in the PCA framework?

Answer: The principal directions (principal components).

21. How are the principal components (PCs) in PCA derived from the SVD of the normalized data matrix?

Answer: PCs = columns of (V) from ($X = U \Sigma V^T$).

22. The square of the singular values (σ^2) from the SVD of the normalized data matrix are proportional to what key property in PCA?

Answer: The variance explained by each principal component.

23. The new data matrix (Z) (the scores) in PCA, which contains the projection of the data onto the PCs, can be obtained using which two SVD component matrices?

Answer: ($Z = U \Sigma$) or equivalently ($Z = X V$).

24. How is the variance captured by each principal component quantified using the singular values (σ)?

Answer: Variance explained by PC (i) = ($\sigma_i^2 / \sum_j \sigma_j^2$).

25. SVD is considered a powerful numerical method for performing PCA, particularly when dealing with what size of datasets?

Answer: Large and/or rectangular datasets where eigendecomposition of the covariance matrix is expensive.

IV. Mathematical Properties (26–30)

26. For a square matrix (A) to be diagonalizable via eigendecomposition, its inverse (Q^{-1}) must exist. What is the property of the eigenvectors that ensures (Q^{-1}) exists?

Answer: The eigenvectors must be linearly independent.

27. For a real symmetric matrix (e.g., Hessian or covariance matrix), what is the relationship between its eigenvector matrix (Q) and its inverse (Q^{-1})?

Answer: ($Q^{-1} = Q^T$) (eigenvectors are orthonormal).

28. What is the relationship between the singular values of (X) and the eigenvalues of the matrix ($X X^H$)?

Answer: Eigenvalues of ($X X^H = \sigma_i^2$) (squares of singular values).

29. In the context of SVD, what is the term for (V^H) (where (V) is the ($p \times p$) unitary matrix)?

Answer: (V^H) is the conjugate transpose (Hermitian) of the right singular vector matrix.

30. The final data matrix (Z) (the PC scores) has columns that are statistically what to each other?

Answer: Uncorrelated (orthogonal) to each other.